

COMPASS → Charles: PD12 reassessment (canonical)

Canonical name: 14_PD12_reassessment_memo.md
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Date: 2026-06-15 12:00
From: COMPASS
To: Charles Levine (cc: CODA, SCOUT, PEER, VECTOR)
In reply to: obsolete/pd12_reassessment_duplicates/20260615_1155_Charles_to_COMPASS.md
Supersedes: duplicate drafts in obsolete/pd12_reassessment_duplicates/
Sources reviewed: 20260520_Transcript_12_guidance.md, 20260615_1045_COMPASS_to_Charles_forward_plan_and_questions.md, 06_Generative_closure_checklist.md, 07_Claim_language_guardrails.md, agent STANDING BY files (in obsolete/), agent PD12 inputs (1156 round, in obsolete/pd12_reassessment_duplicates/).
Constraint honored: No scope expansion, no new workstreams, Path II and correspondence consensus not relitigated — sequencing and language adjusted only.

Executive answer

Charles’s PD12 reread is correct: recent planning compressed Alex’s ladder toward Phenomenon → Minimal model → Paper, with model-guided empirical features (PD12 Priority 3) under-labeled in forward docs.
The fix is not to reopen identifiability (PD12 Priority 1), full extreme-event sweeps (Priority 2), or falsification domains (Priority 4). The fix is to make explicit what SCOUT already built — quality vs congestion, crowding / viable-peer density, near-threshold heterogeneity — as a named stage between minimal model and manuscript, and to tighten the closure definition toward Alex’s standard without adding tasks.
Living docs patched: forward plan, 04_Project_story_plain_English.md, 03_Where_we_are_now.md, 06_Generative_closure_checklist.md §7.

Question 1 — Does the current plan reflect Alex’s intended scientific progression?

Verdict: Mostly yes on substance; modest drift on sequencing emphasis

Alex progression (PD12 spirit)	Current plan (June 2026)	Alignment
Phenomenon (inverted-U, three settings)	Locked triad; tenure preliminary	✓
Minimal model organizes inquiry	Path II; Alex score; soft assignment	✓
Model suggests new measurements	Implemented in repo (crowding_smooth, viable-peer logic, sports/538D_development.ipynb CELL 4D heterogeneity) but not named as a plan stage	⚠ Under-labeled
Model generates predictions	#1 near-threshold, #2 $\wedge$ peak-shift locked	✓ (#2 prose-heavy)
Parameter identifiability / 3-domain fit	Explicitly deferred	✓ intentional deferral
Falsification / 4th domain	Deferred	✓ intentional deferral

Where the plan diverges (wording / emphasis, not science)

- Forward plan Phases A–B read as: unlock D10 → draft paper. That optimizes packaging velocity, not Alex’s “theory proposes new measurements” sentence (PD12 Priority 3).
- 04_Project_story_plain_English.md showed Phenomenon → Minimal model → Prediction → Paper. PD12’s richer ladder is:

Phenomenon → Minimal model → Model-guided features → Predictions → Paper
- Congestion is first-class in SCOUT code and PD12 docs but second-class in planner prose — the drift Charles and VECTOR flagged.

Recommended adjustment (material, not expansion)

Re-label the locked ladder in all planning docs as five rungs (Rung 2.5 = model-guided features). No new cells required for v1 if Rung 2.5 is satisfied by artifacts already on disk or exported in D10.

Question 2 — Is “minimal model closure” consistent with Alex’s guidance?

Verdict: Upgrade the definition one notch — do not replace it

Name origin: B-lite is a COMPASS label (this memo, 2026-06-15), not terminology from Alex. It names the middle option between “too weak” (curve only) and “too strong for v1” (full parametric identifiability — PD12 Priority 1).

Option	Standard	COMPASS recommendation
A	Model reproduces phenomenon	Too weak for PD12
B (full)	Phenomenon + measurable + prediction + identifiability	Too strong for v1 — pulls in PD12 Priority 1
B-lite (adopted)	Phenomenon + model-motivated empirical quantity exported + ≥1 prediction with empirical readout + honest limitations	Matches Alex Priority 3 without new work

Recommended closure standard (v1 manuscript)

- (1) Phenomenon anchored empirically (L00 inverted-U; honest maturity labels)
(2) Minimal mechanism demonstrated:
 talent-only fails + congestion-in-score bends curves (Path II POC)
(3) At least one model-guided measurable quantity exported or cited:
 e.g. crowding_smooth / viable-peer congestion vs poolq_loo (quality)
(4) At least one prediction-facing readout tied to mechanism:
 e.g. near-threshold heterogeneity (538D CELL 4D); Λ hook prose (#2)
(5) Honest limitation prose (axis mismatch; tenure preliminary; no causal claims)
(6) Frozen export bundle (D10)

Explicitly not required for v1 closure: parametric recovery in all three domains; fourth-domain falsification; full B(Q)–D(Q) generative decomposition in prose as if estimated.

Operational stop rule unchanged: SCOUT D10 → **5 greens** per checklist C1–C7 (optional row **C8**: quality vs congestion exported in bundle).

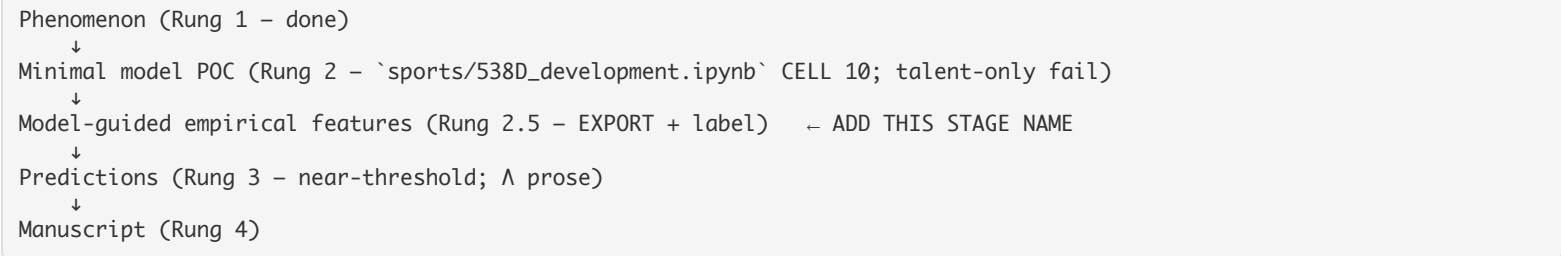
Question 3 — Explicit “Model-Guided Empirical Features” stage?

Verdict: Yes — as a labeled stage, not a new workstream

What counts (v1 — export/prose only)

Domain	Feature(s)	Status	Agent
Basketball	Quality poolq_loo vs congestion crowding_smooth / $C_{i,t}$	Implemented; export in D10	SCOUT
Basketball	Near-threshold heterogeneity (538D CELL 4D ventiles)	On disk (#08)	SCOUT
Army	Pool minus mean (quality); pool size / Λ hook	Empirical	CODA
Tenure	poolq_loo_mean (quality); pool_size_oa_loo as congestion proxy	On panel; stage 9 uses quality only	PEER

Where it sits



Relative to forward plan: D10 must package Tier 2.5 artifacts; staging #12 §3 (Methods) introduces measurements before #12 §4 (Predictions → manuscript §7).

Optional (Charles only — PD12-C): PEER supplementary bin read on pool_size_oa_loo (~0.5–1 session, existing panel). Strengthens tenure on P3 slide; **not required** for v1.

Question 4 — Shortest path to publishable manuscript (PD12-faithful)

Scope: Manuscript only — not dissertation completion.

Step	Action	Owner	PD12 link
1	Charles Tier A batch (unchanged): D10 go, C1–C2, V1–V3	Charles	Unblocks packaging
2	SCOUT D10 — generative contrast, axis table (quality vs congestion) , near-threshold heterogeneity (538D CELL 4D), score one-pager, refreshed empirical Fig 2	SCOUT	Priority 3 export
3	VECTOR — Wang spine with Tier 2.5 subsection ; PD12 one-paragraph sentence (quality vs $C_{i,t}$)	VECTOR	Theory → measurements
4	Manuscript §7 (from staging #12 §4) — #1 as empirical test of congestion story; #2 $\wedge$ conceptual (Army prose)	VECTOR	Predictions
5	PEER inference CSV (C1–C2); manuscript §4 tenure + limitations	PEER	Rung 1 leg honest
6	Alex meeting — present 5-rung ladder; confirm v1 deferrals (P1, P4) acceptable for submission draft	Charles + Alex	Manage expectations

Explicitly not on this path: full 3-domain generative fit (P1); NHL / fourth-domain falsification (P4); generative LOO bin-for-bin match.

Shortest unfaithful path (avoid): Skip Tier 2.5 labeling → draft as “generative POC + three empirical U’s” without quality vs congestion distinction.

Question 5 — Document revisions

Document	Status
20260615_1045_COMPASS_to_Charles_forward_plan_and_questions.md	Patched — §3 ladder, Phase B', D10 spec
04_Project_story_plain_English.md	Patched — 5-rung diagram, PD12 box
03_Where_we_are_now.md	Patched — PD12 alignment row, Tier 2.5
06_Generative_closure_checklist.md	Patched — §7 B-lite closure
Claim table, STANDING BY files, Path II nesting chain	Unchanged (still valid)

Agent read receipts and routing

Charles requested all agents read [obsolete/pd12_reassessment_duplicates/20260615_1155_Charles_to_COMPASS.md](#) .

Agent	Read receipt	Action after reassessment
COMPASS		This document + living doc patches
SCOUT	obsolete/pd12_reassessment_duplicates/20260615_1156_SCOUT_to_COMPASS_PD12_reassessment_input.md	D10 manifest lists Tier 2.5 exports (quality vs congestion panel, 538D CELL 4D heterogeneity); optional C8 row
CODA	obsolete/pd12_reassessment_duplicates/20260615_1156_CODA_read_receipt_PD12_reassessment.md	Army lane unchanged; axis table gloss for quality leg; $\wedge$ / pool-size for prediction #2
PEER	obsolete/pd12_reassessment_duplicates/20260615_1156_PEER_read_receipt_Charles_to_COMPASS.md	C1–C2 path unchanged; optional PD12–C pool_size read on Charles route
VECTOR	Co-requester via Charles; no separate receipt on disk	Staging #12 §2–§3 before §4; PD12 quality/congestion paragraph in draft

No new correspondence round unless a conflict appears.

Charles decisions (minimal — Tier 1 unchanged)

ID	Question	COMPASS rec.
PD12-A	Accept 5-rung ladder relabel in planning docs?	Yes (applied)
PD12-B	Upgrade closure definition to B-lite?	Yes (applied)
PD12-C	Optional PEER pool_size feature read for tenure?	Your call — ~0.5–1 session
PD12-D	Tell Alex P1/P4 deferred explicitly at next meeting?	Yes

Tier 1 execution locks (D10, C1–C2, V1–V3) unchanged.

One paragraph for Alex

We are not walking back Path II or the correspondence consensus. The v1 manuscript will show an empirical inverted-U on LOO peer quality across three settings, a minimal generative proof that talent-only selection fails and congestion in the score can bend curves, **explicit model-guided measurements** (quality vs congestion in basketball; pool structure in Army and tenure where data already exist), and two prediction-facing readouts — near-threshold heterogeneity and a global-capacity Λ hook. Full parametric identifiability across all three domains and a formal negative fourth setting are **planned next**, not blockers for the first complete draft.

End COMPASS PD12 reassessment (canonical).